

Supplying a gate operator with the correct electrical service is crucial to the performance of the operator and the life of its electrical components. If the wire size used is too small, the voltage loss, especially during motor startup, will prevent the motor from attaining its rated horsepower. The percentage of horsepower lost is far greater than the percentage of voltage loss.

A voltage loss can also cause the control components to chatter while the motor is starting, substantially reducing their life due to the resultant arcing. There is no way to restore lost performance resulting from undersized wires, except to replace them. To avoid costly rewiring, be sure to choose a sufficient wire size during site planning phases and at initial installation.

NOTICE

DO NOT use undersized power wires. This can lead to power supply brown outs, SmartTouch 720/725 Controller resets, and premature motor failure.

The tables on the following page are based on copper wire and allow for a 5% voltage drop. The ampere values shown are the service factor ampere rating of the motor (maximum full load at continuous duty). At minimum, a 20A circuit (protected with a 20A Inverse time Breaker) should be provided depending on rated voltage of operator.

Always connect electrical power and ground the operator in accordance with the NFPA 780 & NEC, Article 430 and Article 250. Research and adhere to other local codes that may apply.

LOW VOLTAGE CONTROL WIRING

The SmartTouch 720/725 Controller has very sensitive control inputs. The following is a chart of maximum distances for wire size:

Wire Size	Maximum Distance
18 ga	7.0 miles (11 km)
20 ga	3.5 miles (5.6 km)
22 ga	2.7 miles (4.3 km)
24 ga	2.0 miles (3.2 km)
26 ga	1.0 mile (1.6 km)
28 ga	3700 feet (1.1 km)

WIRE SIZING AND RUNS

SLIDEDRIVER II WIRING CHARTS (INCOMING POWER)

The maximum distance shown is from the operator to the power source, assuming that source power is from a panel box with adequate capacity to support the addition of this motor load. The values are for one operator, with no other loads applied to the branch circuit. Avoid placing more than one operator to a circuit, but if you must, be certain to reduce the maximum allowed wire distance by half.

Make sure proper wiring is being used. The following table shows the maximum allowable wire run from the power source to the operator for various wire sizes. Performance of SD50F-series operators on 1Ø and 3Ø, 50 or 60 Hz Power.

Note: Distance shown in U.S. Standard "feet." Metric equivalent shown in parentheses.

Table 4. SlideDriver II Wire Size Chart – SD15 and SD40						
Phase Ø	1	1	1	3	3	3
Voltage	115	208	230	208	230	460
Horsepower (Hp)	1	1	1	1	1	1
Maximum Rated Current Draw (A)	15.6	8.8	8.5	6.3	6.1	3.7
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	35 (11 m)	115 (35 m)	125 (38 m)	300 (91 m)	340 (104 m)	1135 (346 m)
10	55 (17 m)	185 (56 m)	205 (62 m)	475 (145 m)	545 (166 m)	1800 (549 m)
8	90 (27 m)	290 (88 m)	320 (98 m)	750 (229 m)	860 (262 m)	2835 (864 m)
6	140 (43 m)	450 (137 m)	500 (152 m)	1170 (357 m)	1340 (408 m)	4415 (1346 m)
4	220 (67 m)	705 (215 m)	780 (238 m)	1825 (556 m)	2085 (636 m)	6875 (2096 m)
2	335 (102 m)	1085 (331 m)	1200 (366 m)	2800 (853 m)	3200 (975 m)	10555 (3217 m)

Table 5. SlideDriver II Wire Size Chart – SD50F and SD80V						
Phase Ø	1	1	3	3	3	3
Voltage (V)	208	230	208	230	380	460
Horsepower (Hp)	2	2	2	2	2	2
Maximum Rated Current Draw (A)	19.9	17.8	13.7	12	8.7	8.1
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	150 (46 m)	175 (53 m)	230 (70 m)	295 (90 m)	675 (206 m)	870 (265 m)
10	240 (73 m)	280 (85 m)	370 (113 m)	470 (143 m)	1070 (326 m)	1385 (422 m)
8	380 (116 m)	440 (134 m)	585 (178 m)	745 (227 m)	1685 (514 m)	2180 (664 m)
6	590 (180 m)	685 (209 m)	910 (277 m)	1160 (354 m)	2625 (800 m)	3390 (1033 m)
4	915 (279 m)	1070 (326 m)	1415 (431 m)	1805 (550 m)	4080 (1240 m)	5275 (1608 m)
2	1405 (428 m)	1640 (500 m)	2165 (660 m)	2765 (843 m)	6255 (1907 m)	8085 (2464 m)

200V table on next page.

Table 6. SlideDriver II Wire Size Chart – SD200V

Phase Ø	3	3	3	3
Voltage (V)	208	230	380	460
Horsepower (Hp)	5	5	5	5
Maximum Rated Current Draw (A)	26	22.7	15.6	13.2
Wire Gauge	Distance	Distance	Distance	Distance
12	57 (18 m)	73 (22 m)	175 (53 m)	253 (77 m)
10	90 (27 m)	115 (35 m)	280 (85 m)	400 (122 m)
8	145 (44 m)	183 (56 m)	440 (134 m)	633 (193 m)
6	225 (69 m)	285 (87 m)	688 (210 m)	985 (300 m)
4	350 (107 m)	445 (136 m)	1070 (326 m)	1533 (467 m)
2	540 (165 m)	683 (208 m)	1645 (501 m)	2353 (717 m)